In Silico Mapping of Visual Categorical Selectivity Across the Whole Brain

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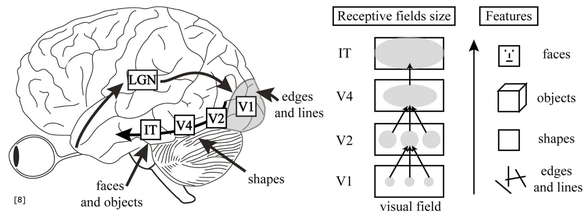

- Extensive neuroimaging experiments, especially functional magnetic resonance imaging (fMRI), have mapped a few prominent categories, faces, places, words, bodies, and food , to dedicated brain regions.
- But visual perception goes beyond simple categories and it remains an open question what higher-level visual concepts enable humans to make sense of the complex world.
- Here, we leverage recent advances in artificial intelligence and a large-scale fMRI dataset to explore the visual selectivity of brain regions beyond the well-studied visual cortex.
- We found many parcels with more complex selectivity transcending simple categorical concepts.
- Finally, we verify our labels by testing how well CLIP representations of our "superstimuli" can predict ground-truth activation on a held-out set.

Schaefer functional parcellation

We partitioned the 327,684 cortical vertices across the whole brain into 1,000 functionally homogeneous parcels using the Schaefer resting-state functional connectivity parcellation [9].

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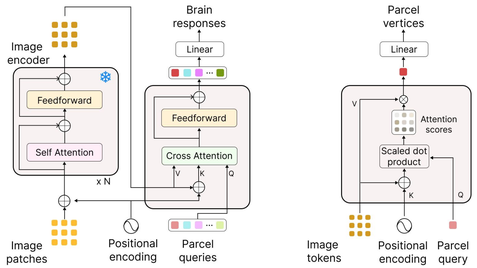
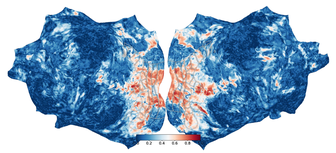
Transformer Brain encoder



Transformer Brain encoder (TBEn); Adeli et al (2025)[5]

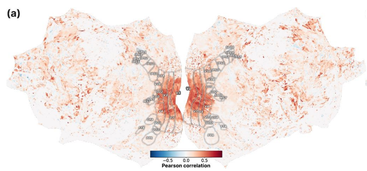
We built a transformer based brain encoder [5] to predict brain activity from visual features [3]. We trained the model on the NSD dataset [1], which includes up to 10,000 image-fMRI pairs for each of 8 subjects.

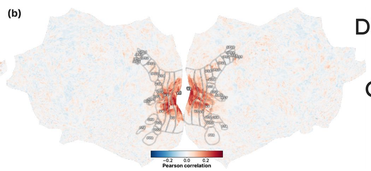
Prediction accuracy (pearson correlation)



As expected, our model performs well on predicting activity in the visual cortex. Several areas beyond the visual cortex are also well-predicted, which may offer insight into downstream visual processing.

TBEn vs Linear

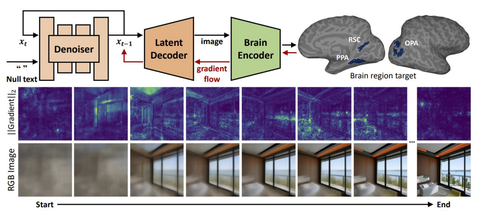




Dinov2 vs CLIP

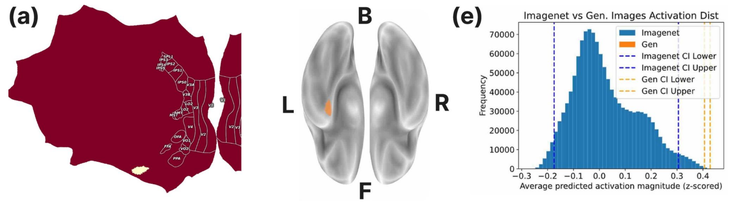
Generating images with diffusion

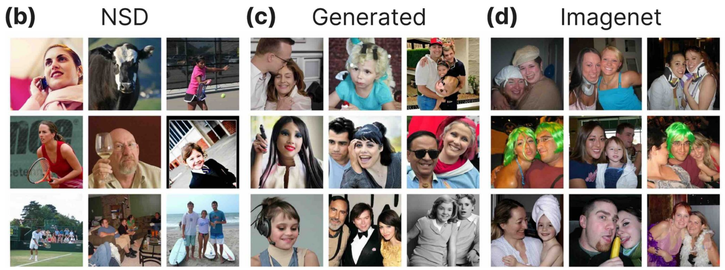
We used a diffusion based model, BrainDIVE [2], to generate images that would maximally activate a given parcel. The encoder model serves as a "digital twin" that is fully observable, upon which we perform extensive experimentation to better understand selectivity beyond the visual cortex.



BrainDiVE; Luo et al (2023)

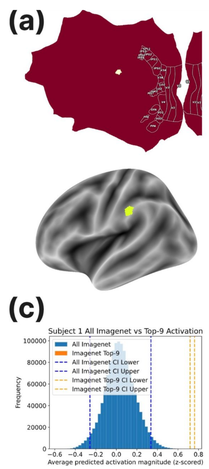
Sanity check: selectivity of known regions (aTL-faces)

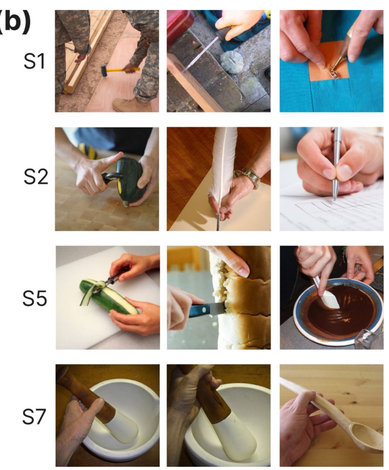




We start by demonstrating the selectivity of a parcel that overlaps significantly with a known region (a), aTL-faces, which is known to respond to faces [7]. We chose images from the NSD, BrainDIVE, and Imagenet [4] that maximally activate the parcel (b).

Demonstrating complex selectivity





"...utensils in action, often in practical, hands-on settings..."

"...hands actively engaging in a step of making food..."

"...objects or body parts being used as writing instruments or tools..."

"...action involving tools or sports equipment..."